

DEVELOPMENT OF AN ARTIFICIAL INTELLIGENCE-BASED PHISHING DETECTION SYSTEM TO ENHANCE CYBER RESILIENCE IN THE DIGITAL ERA

Prepared By:

Tequel Team: Riki Ardiansyah¹, Jovan Akbar Setiodono², Ar'rasya Putra³

Universitas Teknokrat Indonesia^{1,2,3}

Riki_ardiansyah@teknokrat.ac.id¹, jovan_akbar_setiodono@teknokrat.ac.id²,
arrasya_putra@teknokrat.ac.id³

ABSTRACT

The rapid development of digital technology has brought significant benefits to society; however, it has also increased the risk of cyber threats, particularly phishing attacks. Phishing is one of the most commonly used methods by cybercriminals to steal sensitive information such as personal data, passwords, and financial information. This study aims to develop an Artificial Intelligence (AI)-based phishing detection system capable of automatically identifying malicious websites or links. The methodology includes phishing dataset collection, machine learning model training, and system performance evaluation based on accuracy, precision, and recall metrics. The expected outcome is a system that enhances internet user security and supports cyber resilience in the digital era.

Keywords: Cyber Security, Cyber Resilience, Phishing, Artificial Intelligence, Machine Learning.

CHAPTER 1 INTRODUCTION

Digital transformation has significantly reshaped modern society by improving accessibility, connectivity, and efficiency across numerous sectors. While these technological advancements offer substantial benefits, they have also introduced increasingly sophisticated cybersecurity challenges. Among these threats, phishing attacks continue to be one of the leading causes of data breaches and financial losses worldwide. According to recent cybersecurity reports, phishing remains a preferred attack method due to its simplicity, effectiveness, and ability to exploit human vulnerabilities.

Many internet users still experience difficulties in distinguishing legitimate websites from fraudulent ones. Cybercriminals take advantage of this lack of awareness by creating convincing fake websites, emails, and messages that encourage victims to disclose sensitive information. The emergence of new forms of cyber threats highlights the necessity for individuals and organizations to enhance their cybersecurity preparedness and resilience.

Cyber resilience emphasizes the capability of a system to anticipate, withstand, detect, respond to, and recover from cyber incidents. In this context, Artificial Intelligence offers promising opportunities to strengthen cybersecurity measures through automated threat detection and analysis. Therefore, this research proposes the development of an AI-based phishing detection system designed to identify malicious websites and suspicious links efficiently and accurately. The main objective of this study is to develop a phishing detection system utilizing machine learning techniques. Furthermore, the study seeks to evaluate the effectiveness of the proposed system in identifying phishing threats and examine its contribution to improving cyber resilience. The outcomes of this research are expected to provide academic value for future studies, practical benefits for internet users, and broader social impacts by increasing awareness of digital security issues.

CHAPTER 2 LITERATURE REVIEW

Cyber resilience refers to the ability of organizations and individuals to maintain operational continuity despite experiencing cyberattacks or digital disruptions. The concept extends beyond prevention by incorporating detection, response, recovery, and adaptation capabilities that ensure sustainable cybersecurity practices.

Phishing is a form of cyberattack that aims to obtain confidential information by impersonating trusted entities. Attackers commonly distribute fraudulent emails, websites, or messages that resemble legitimate sources in order to deceive users into revealing sensitive data. As phishing techniques continue to evolve, traditional detection methods often struggle to identify newly emerging threats.

Artificial Intelligence is a technology that enables computer systems to perform tasks requiring human intelligence, including learning, reasoning, pattern recognition, and decision-making. In cybersecurity, AI has become an essential tool for identifying anomalies, detecting threats, and automating security processes.

Machine Learning, a subset of Artificial Intelligence, allows systems to learn from historical data and improve their performance over time without explicit programming. By analyzing large datasets, machine learning algorithms can identify patterns associated with phishing attacks and generate predictive models capable of classifying suspicious websites and links with high accuracy.

CHAPTER 3 RESEARCH METHODOLOGY

This research adopts a quantitative and experimental approach to develop and evaluate an AI-based phishing detection system. The study begins with a comprehensive literature review of cybersecurity, phishing detection techniques, artificial intelligence, and machine learning applications in digital security.

The data used in this study consist of phishing URLs and legitimate website URLs collected from publicly available datasets. After data collection, preprocessing activities are conducted to improve data quality through cleaning, duplicate removal, and normalization processes.

The prepared dataset is then utilized for model training. Several machine learning algorithms, including Random Forest, Decision Tree, and Logistic Regression, are employed to determine the most effective classification model. Following the training phase, the system undergoes performance evaluation using standard metrics such as accuracy, precision, recall, and F1-score. The results are analyzed to assess the capability of the proposed system in detecting phishing threats and supporting cyber resilience.

CHAPTER 4 SYSTEM DESIGN

The proposed system is designed to analyze website URLs and links obtained from emails or instant messaging platforms. Once a URL is submitted, the system performs structural analysis and extracts relevant features associated with phishing characteristics. These features are then processed by the trained machine learning model to generate predictions regarding the legitimacy of the website.

The system produces results in the form of a security classification indicating whether the URL is safe or potentially dangerous. Additionally, the system provides a risk score expressed as a percentage to help users better understand the level of threat associated with the analyzed link. Through this approach, users can receive immediate warnings before accessing suspicious websites.

CHAPTER 5 RESEACH SCHEDULE

Activities	Week 1	Week 2	Week 3	Week 4
Literature Review	✓			

Data Collection		✓	✓	
Systems Development		✓	✓	
System testing			✓	
Report writing				✓

CHAPTER 6 BUDGET PLAN

No	Requirements	Cost
1	Internet and data	Rp 100,000
2	System hosting	Rp 150,000
3	Documentation	Rp 50,000
4	Team operational costs	Rp 100,000
Total		Rp 400,000

CHAPTER 7 CONCLUSION

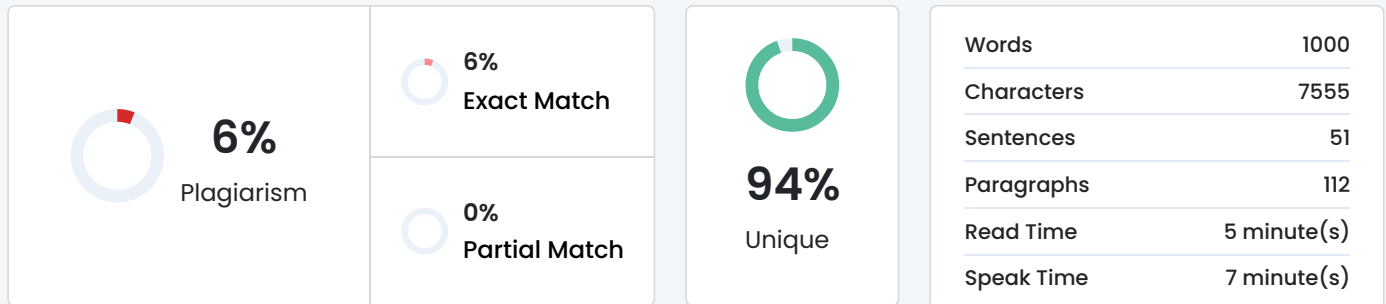
Phishing attacks continue to pose significant challenges to cybersecurity in the digital era. The increasing sophistication of cybercriminal techniques requires innovative solutions capable of detecting threats quickly and accurately. The implementation of Artificial Intelligence and Machine Learning technologies offers a promising approach to addressing this issue through automated phishing detection mechanisms.

The proposed AI-based phishing detection system is expected to improve cybersecurity awareness, reduce the risk of sensitive data theft, and contribute to the enhancement of cyber resilience. Future developments may focus on integrating the system into web browsers and mobile applications, enabling broader adoption and providing real-time protection for internet users.

REFERENCES

- Afifah, E. F. N., Simatangkir, D. W. E., & Faliha, N. S. (2025). *Cybersecurity in banking: Challenges and solutions in the digital era*. Jurnal Manajemen Informatika dan Administrasi, 2(1).
- Hartanto, M. B., Winarko, T., Yunita, H. D., Fahurian, F., Yuniarthe, Y., & Zuhri, K. (2025). *Cybersecurity prediction model using artificial intelligence for digital threat mitigation*. Prosiding Konstelasi, 2(1).
- Karunia, W. A., Zahra, A. F., & Amrozi, Y. (2025). *Evaluation of emerging threats in information security: A systematic literature review of cybersecurity vulnerabilities in the post-pandemic era*. Cyber Security dan Forensik Digital, 8(1).
- Rinanto, D. R., & Nurmiati, E. (2026). *Analysis of banking information system security capabilities in facing the evolution of cybercrime*. Jurnal Sains dan Sistem Riset, 4(3).

Plagiarism Scan Report



Content Checked For Plagiarism

DEVELOPMENT OF AN ARTIFICIAL INTELLIGENCE-
BASED PHISHING DETECTION SYSTEM TO ENHANCE
CYBER RESILIENCE IN THE DIGITAL ERA

Prepared By:

Tequel Team: Riki Ardiansyah¹, Jovan Akbar Setiodono², Ar'rasya Putra³

Universitas Teknokrat Indonesia^{1,2,3}

Riki_ardiansyah@teknokrat.ac.id¹, jovan_akbar_setiodono@teknokrat.ac.id²,

arrasya_putra@teknokrat.ac.id³

ABSTRACT

The rapid development of digital technology has brought significant benefits to society; however, it has also increased the risk of cyber threats, particularly phishing attacks. Phishing is one of the most commonly used methods by cybercriminals to steal sensitive information such as personal data, passwords, and financial information. This study aims to develop an Artificial Intelligence (AI)-based phishing detection system capable of automatically identifying malicious websites or links. The methodology includes phishing dataset collection, machine learning model training, and system performance evaluation based on accuracy, precision, and recall metrics. The expected outcome is a system that enhances internet user security and supports cyber resilience in the digital era.

Keywords: Cyber Security, Cyber Resilience, Phishing, Artificial Intelligence, Machine Learning.

CHAPTER 1 INTRODUCTION

Digital transformation has significantly reshaped modern society by improving accessibility, connectivity, and efficiency across numerous sectors. While these technological advancements offer substantial benefits, they have also introduced increasingly sophisticated cybersecurity challenges. Among these threats, phishing attacks continue to be one of the leading causes of data breaches and financial losses worldwide. According to recent cybersecurity reports, phishing remains a preferred attack method due to its simplicity, effectiveness, and ability to exploit human vulnerabilities.

Many internet users still experience difficulties in distinguishing legitimate websites from fraudulent ones. Cybercriminals take advantage of this lack of awareness by creating convincing fake websites, emails, and messages that encourage victims to disclose sensitive information. The emergence of new forms of cyber threats highlights the necessity for individuals and organizations to enhance their cybersecurity preparedness and resilience.

Cyber resilience emphasizes the capability of a system to anticipate, withstand, detect, respond to, and recover from cyber incidents. In this context, Artificial Intelligence offers promising opportunities to strengthen cybersecurity measures through automated threat detection and analysis. Therefore, this research proposes the development of an AI-based phishing detection system designed to identify malicious websites and suspicious links efficiently and accurately. The main objective of this study is to develop a phishing detection system utilizing machine learning techniques. Furthermore, the study seeks to evaluate the effectiveness of the proposed system in identifying phishing threats and examine its contribution to improving cyber resilience. The outcomes of this research are expected to provide academic value for future studies, practical benefits for internet users, and broader social impacts by increasing awareness of digital security issues.

CHAPTER 2 LITERATURE REVIEW

Cyber resilience refers to the ability of organizations and individuals to maintain operational continuity despite experiencing cyberattacks or digital disruptions. The concept extends beyond prevention by incorporating detection, response, recovery, and adaptation capabilities that ensure sustainable cybersecurity practices.

Phishing is a form of cyberattack that aims to obtain confidential information by impersonating trusted entities. Attackers commonly distribute fraudulent emails, websites, or messages that resemble legitimate sources in order to deceive users into revealing sensitive data. As phishing techniques continue to evolve, traditional detection methods often struggle to identify newly emerging threats.

Artificial Intelligence is a technology that enables computer systems to perform tasks requiring human intelligence, including learning, reasoning, pattern recognition, and decision-making. In cybersecurity, AI has become an essential tool for identifying anomalies, detecting threats, and automating security processes.

Machine Learning, a subset of Artificial Intelligence, allows systems to learn from historical data and improve their performance over time without explicit programming. By analyzing large datasets, machine learning algorithms can identify patterns associated with phishing attacks and generate predictive models capable of classifying suspicious websites and links with high accuracy.

CHAPTER 3 RESEARCH METHODOLOGY

This research adopts a quantitative and experimental approach to develop and evaluate an AI-based phishing detection system. The study begins with a comprehensive literature review of cybersecurity, phishing detection techniques, artificial intelligence, and machine learning applications in digital security.

The data used in this study consist of phishing URLs and legitimate website URLs collected from publicly available datasets. After data collection, preprocessing activities are conducted to improve data quality through cleaning, duplicate removal, and normalization processes. The prepared dataset is then utilized for model training. Several machine learning algorithms, including Random Forest, Decision Tree, and Logistic Regression, are employed to determine the most effective classification model. Following the training phase, the system undergoes performance evaluation using standard metrics such as accuracy, precision, recall, and F1-score. The results are analyzed to assess the capability of the proposed system in detecting phishing threats and supporting cyber resilience.

CHAPTER 4 SYSTEM DESIGN

The proposed system is designed to analyze website URLs and links obtained from emails or instant messaging platforms. Once a URL is submitted, the system performs structural analysis and extracts relevant features associated with phishing characteristics. These features are then processed by the trained machine learning model to generate predictions regarding the legitimacy of the website.

The system produces results in the form of a security classification indicating whether the URL

is safe or potentially dangerous. Additionally, the system provides a risk score expressed as a percentage to help users better understand the level of threat associated with the analyzed link. Through this approach, users can receive immediate warnings before accessing suspicious websites.

CHAPTER 5 RESEACH SCHEDULE

Activities Week 1 Week 2 Week 3 Week 4

Literature Review ✓

Data Collection ✓ ✓

Systems

Development

✓ ✓

System testing ✓

Report writing ✓

CHAPTER 6 BUDGET PLAN

No Requirements Cost

1 Internet and data Rp 100,000

2 System hosting Rp 150,000

3 Documentation Rp 50,000

4 Team operational costs Rp 100,000

Total Rp 400,000

CHAPTER 7 CONCLUSION

Phishing attacks continue to pose significant challenges to cybersecurity in the digital era. The increasing sophistication of cybercriminal techniques requires innovative solutions capable of detecting threats quickly and accurately. The implementation of Artificial Intelligence and Machine Learning technologies offers a promising approach to addressing this issue through automated phishing detection mechanisms.

The proposed AI-based phishing detection system is expected to improve cybersecurity awareness, reduce the

Matched Source

Similarity 1%

Title: [Award | SBIR](#)

Award | SBIR | SBIR (.gov)<https://www.sbir.gov> › awards | SBIR (.gov)<https://www.sbir.gov> › awards
<https://www.sbir.gov/awards/68025>

Similarity 1%

Title: [AI In Cybersecurity: Key Benefits, Defense Strategies, & Future ...](#)

AI In Cybersecurity: Key Benefits, Defense Strategies, & Future ... Fortinet<https://www.fortinet.com> › resources › cyberglossary
Fortinet<https://www.fortinet.com> › resources › cyberglossary
<https://www.fortinet.com/resources/cyberglossary/artificial-intelligence-in-cybersecurity>

Similarity 1%

Title: [Bixby Developer](#)

If a concept extends an enum , it must also be an enum . If the concept extends multiple enum concepts, all the parents must have identical symbol sets. Example.

<https://bixbydevelopers.com/dev/docs/reference/type/structure.extends>

Similarity 1%

Title: [What is Machine Learning? How AI Learns from Data](#)

· 4 days ago · Machine learning is a branch of artificial intelligence that enables computers to learn from data and improve their performance without explicit programming. Discover how ML works, its key concepts, and real-world applications.

<https://www.exp.science/education/what-is-machine-learning-how-ai-learns-from-data>
